FIITJEE

JEE (Advanced)-2019

ALL INDIA TEST SERIES

PART TEST – I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*

   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*

   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*

   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST - I*** *and* ***LIST - II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST - I*** *and* ***LIST - II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):

(SECTION – D)

For questions **11 to 18, 29 to 36, 47 to 54.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

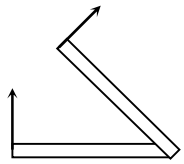
Physics

PART – I

SECTION – A
(One or More than one correct type)

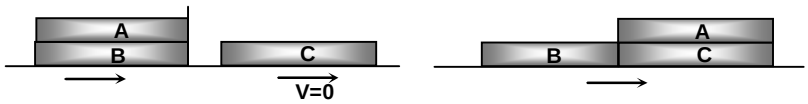
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A long thin uniform rod lies flat on the table as shown. One end of the rod is slowly pulled up by a force that remains perpendicular to the rod at all times. The rod is to be brought to the vertical position without any slipping of the bottom end? The coefficient of static friction between rod and ground can be:



(A) 0.2 (B) 0.3
(C) 0.4 (D) 0.5

2. Board A is placed on board B as shown. Both boards slide, without moving with respect to each other, along a frictionless horizontal surface at a speed v . Board B hits a resting board C "head-on."



Before Collision **After Collision**

After the collision, boards B and C move together, and board A slides on top of board C and stops its motion relative to C in the position shown on the diagram. All three boards have the same mass, size, and shape. It is known there is no friction between boards A and B; the coefficient of kinetic friction between boards A and C is μ_k.

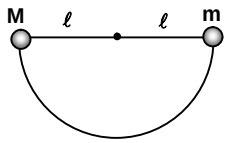
(A) The length of each board is $\frac{v^2}{6\mu_k g}$

(B) The final speed of each board is 2v/3.

(C) Total work done by frictional force on the system is zero.

(D) Total work done by frictional force on C is $\frac{mv^2}{12}$.

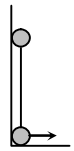
3. Two balls of masses M and m are attached to two threads of length l, such that the two threads are suspended at the same point. The balls are both deflected to the horizontal position of the threads, released from rest at the same time, and they collide head on. (The collision is perfectly elastic.). If the ball of mass m reaches the maximum height which is allowed M/m can be :



(A) 0.5 (B) 1.5
(C) 2 (D) 2.5

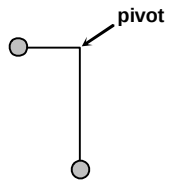
Space for Rough work

4. A dumbbell consists of a light rod of length r and two small masses m attached to it. The dumbbell stands vertically in the corner formed by two frictionless planes. After the bottom end is slightly moved to the right, the dumbbell begins to slide. At the moment the top end loses contact with the vertical plane:



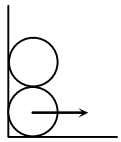
(A) The acceleration of top mass is g.
(B) Horizontal component of acceleration of bottom mass is zero.
(C) The height of the top mass will be 2r/3.
(D) Speed of the bottom mass will be $\frac{2}{3}\sqrt{gR}$.

5. Two small spheres of mass m each are attached to the ends of a light rigid rod bent at a right angle and pivoted in the vertical plane at the vertex of the right angle. The vertical part of the rod is three times as long as the horizontal part. Initially, the rod is held at rest in the position shown in the diagram. The pivot is frictionless. Assume that mass of rods are negligible. Immediately after the system is released.



(A) Horizontal component of acceleration of center of mass of the system is $\frac{3}{10}$ m/s^2 towards right.
(B) Vertical component of acceleration of center of mass of the system is $\frac{1}{20}$ m/s^2 downwards.
(C) Horizontal component of hinge reaction is $\frac{3mg}{10}$ towards right.
(D) Vertical component of hinge reaction is $\frac{19mg}{20}$ upwards.

6. Two identical uniform cylinders of radius R each are placed on top of each other next to a wall as shown. After a disturbance, the bottom cylinder slightly moves to the right and the system comes into motion. Neglect friction between all surfaces.



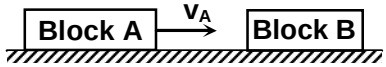
(A) The top cylinder will have maximum acceleration g.
(B) The speed of the bottom cylinder will first increase then decrease.
(C) Minimum acceleration of the bottom cylinder is zero.
(D) Maximum speed of the bottom cylinder is $\frac{2}{3}\sqrt{gR}$

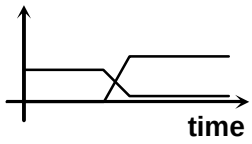
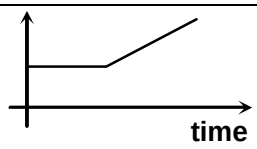
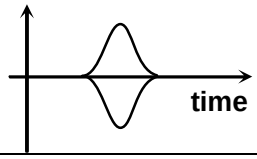
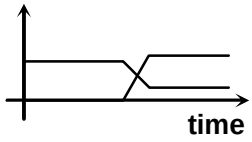
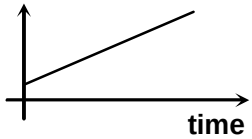
Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

7. Block a moving on frictionless horizontal plane collides head-on with block B initially at rest. The collision is NOT ($0 < e < 1$) perfectly elastic. Match the List-I with the related graph in List-II.

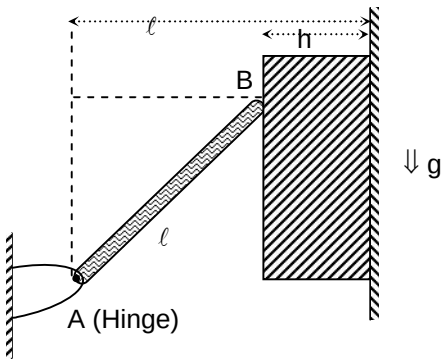


LIST–I		LIST–II	
P.	Forces exerted by the blocks	1.	
Q.	The position of mass centre of the two block system	2.	
R.	Velocity of blocks	3.	
S.	Position of block B	4.	
		5.	

(A) **P → 4; Q → 4; R → 1; S → 4**
(B) **P → 3; Q → 3; R → 1; S → 3**
(C) **P → 3; Q → 5; R → 4; S → 2**
(D) **P → 4; Q → 3; R → 4; S → 1**

Space for Rough work

8. A rectangular block of mass m = 0.3 kg is held stationary with the help of massless rod of length ℓ = 10 m in vertical plane as shown in the figure. Vertical wall is frictionless and coefficient of friction between block and rod is μ = 0.75. Now match the value of physical quantities mentioned in List – I with List – II.

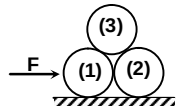


LIST–I		LIST–II (values of a)	
P.	The value of h in meter so that block remains stationary	1.	1.5
Q.	Magnitude of normal force in Newton given by wall to the block	2.	5
R.	Magnitude of Hinge reaction in Newton	3.	4
S.	Magnitude of friction force acting on the block	4.	3
		5.	2

(A) **P → 1, 5; Q → 3; R → 2; S → 4**
(B) **P → 1, 5; Q → 1; R → 2; S → 3**
(C) **P → 3, 1; Q → 2; R → 3; S → 4**
(D) **P → 3, 3; Q → 1; R → 3; S → 4**

Space for Rough work

9. Three identical cylinders are arranged in a triangle as shown in Figure, with the bottom two lying on the ground. The ground and the cylinders are frictionless. You apply a constant horizontal force (directed to the right) on the left cylinder. Let a be the acceleration you give to the system.



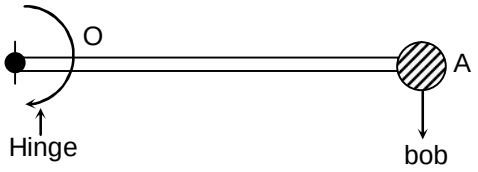
N_{12} is normal reaction between (1) and (2)
N_{13} is normal reaction between (1) and (3)
N_{23} is normal reaction between (2) and (3)
List –II shows various values of a.
Match the functions in List-I with the related charge distributions in List-II.

LIST–I		LIST–II (values of a)	
P.	$N_{12} = 0, N_{13} \neq 0$	1.	g
Q.	$N_{12} \neq 0, N_{13} \neq 0$	2.	$\frac{g}{2\sqrt{3}}$
R.	$N_{12} \neq 0, N_{13} = 0$	3.	$\frac{g}{4\sqrt{3}}$
S.	$N_{23} = 0, N_{13} \neq 0$	4.	$\frac{g}{6\sqrt{3}}$
		5.	Never possible

(A) **P → 3, 4; Q → 2; R → 5; S → 1**
(B) **P → 3, 2; Q → 1; R → 4; S → 3**
(C) **P → 3, 1; Q → 2; R → 3; S → 4**
(D) **P → 3, 3; Q → 1; R → 3; S → 4**

Space for Rough work

10. A uniform rod of mass 6 kg and length 1m is hinged at O and a small bob of mass 1 kg is attached to its other end. The system is kept horizontal and then released. The quantities mentioned under Column I are measured and their magnitudes (in SI) are given in column II, but not necessarily in the proper order. Match the quantities in List I with their corresponding values in List II. Take g = 10 m/s^2.



LIST–I		LIST–II (in SI units)	
P.	Angular acceleration of rod	1.	$\frac{10}{3}$
Q.	Linear acceleration of bob	2.	$\frac{20}{3}$
R.	Force exerted by the rod on the bob	3.	$\frac{40}{3}$
S.	Force exerted by the hinge on the rod	4.	$\frac{50}{3}$
		5.	$\frac{160}{3}$

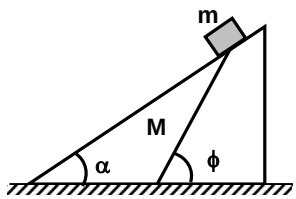
(A) **P → 5; Q → 3, 4; R → 1; S → 2**
(B) **P → 3; Q → 3; R → 1, 4; S → 5**
(C) **P → 5; Q → 3; R → 1, 2; S → 4**
(D) **P → 4; Q → 2, 3; R → 1; S → 5**

Space for Rough work

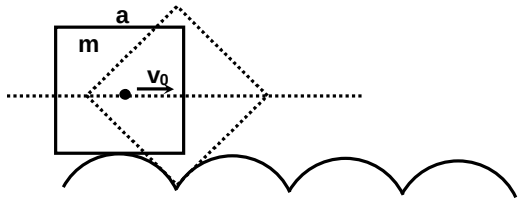
SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. The wedge shown in the figure can slide without friction on a horizontal tabletop. The mass of the wedge is M and its angle of elevation is $\alpha = 30^o$. A body of mass m slides down without friction along the wedge, the path of the body makes an angle of $\varphi = 60^o$ with the ground. Find the ratio of the masses, m/M.



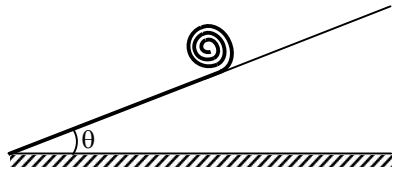
12. A path is made to a uniform density cube of mass m, and of edge a, such that when the cube rolls along the path without skidding, its centre moves along a straight line. At the top of the path a horizontal velocity of center of mass is v_0 and the cube is in pure rolling. The static frictional force is big enough, so the cube do not slide anywhere. The speed of the centre of mass when the cube touches the lowest point of the path is $\frac{\sqrt{k}v_o}{2}$? (The moment of inertia of the cube with respect to its centre of mass is $ma^2/6$). Find k.



13. A block is placed on a long and wide inclined plane that makes angle $\theta = 45°$ with the horizontal. The coefficient of kinetic friction between the block and the plane is $\mu_k = 2$. After a quick push, the block acquires velocity $v_0 = 6\sqrt{2}$ m/s that makes angle $\alpha = 60°$ with the line of fastest descent. Find the time interval t during which the block is in motion. (take g = 10 m/s^2)

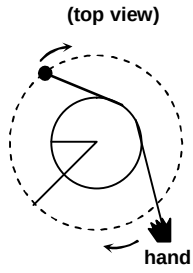
Space for Rough work

14. A carpet runner is placed onto a long straight inclined plane parallel to the steepest line in the plane, from the bottom of the slope till its top. The carpet is thin and flexible, and it cannot slide down because of the friction, but it can easily be lifted, it does not stick to the plane. The top end of the carpet is rolled, a bit, and this hoop of carpet is released. The hoop rolls down the slope with greater and greater speed, while its diameter increases, and finally it reaches the bottom of the slope during a time of t_1. If a solid cylinder is released at the top of the slope, it reaches the bottom in a time of t_2. The ratio of $\frac{t_1}{t_2} = \sqrt{k}$. Find k .



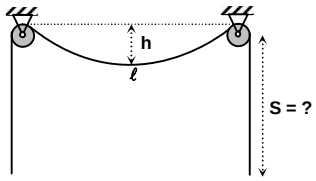
15. A ball is thrown at speed v from zero height on level ground. The angle at which the particle is thrown is such that the area under the trajectory is maximum? If the angle of projection in □/k . Find k .

16. A mass, which is free to move on a horizontal frictionless surface, is attached to one end of a massless string that wraps partially around a frictionless vertical pole of radius r (see the top view in Figure). You hold on to the other end of the string. At t = 0, the mass has speed v_0 = 1m/s in the tangential direction along the dotted circle of radius R shown. Your task is to pull on the string so that the mass keeps moving along the dotted circle. You are required to do this in such a way that the string remains in contact with the pole at all times. (You will have to move your hand around the pole, of course.) What is the speed of the mass at time $t = \frac{R}{2v_0 \tan\theta}$?



17. A mass M collides elastically with a stationary mass m. If M = 2m, then there is a maximal angle of deflection of M. This maximal angle equals ϕ/k radians. Find k .

18. A uniform flexible rope passes over two small frictionless pulleys mounted at the same height (see figure). The length of rope between the pulleys is ℓ = 2√3 m, and its 'sag' is h = 1 m. In equilibrium, what is the length s of the rope segments that hang down on either side?



Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. For the equilibrium $SO_2(g) + Cl_2(g) \rightleftharpoons SO_2Cl_2(g)$, the correct statement(s) is/are

(A) Addition of SO_2(g) at constant volume at equilibrium increases the concentration of SO_2Cl_2(g)
(B) Addition of SO_2(g) at constant volume at equilibrium increases the concentration of SO_2(g)
(C) Addition of SO_2Cl_2(g) at constant volume at equilibrium increases the concentration of SO_2Cl_2(g)
(D) Addition of inert gas at constant volume at equilibrium increases the concentration of SO_2Cl_2 (g)

20. The incorrect statement(s) among the following is/are
(A) 2-Ethylantraquinone on oxidation in air produces H_2O_2.
(B) Peroxydisulphuric acid on hydrolysis produces H_2O_2.
(C) Acetanilide catalyses the decomposition of H_2O_2.
(D) H_2O_2 reduces sodium hypochlorite to sodium chloride.

21. The correct statement(s) among the following is/are
(A) The shape of SF_3Cl molecule is tetrahedral
(B) In PCl_2F_3 all the three F atoms remain in the equatorial position.
(C) The shape of ICl_4^- is square planar.
(D) I_2Cl_6 is a planar molecule in the solid state.

22. The product(s) of hydrolysis of $B_3N_3H_6$ is/are
(A) NH_3 (B) H_3BO_3
(C) H_2 (D) B_2H_6

23. Cyclic silicate among the following is/are
(A) $Ca_3Si_3O_9$ (B) Mg_2SiO_4
(C) $Be_3Al_2Si_6O_{18}$ (D) $Sc_2Si_2O_7$

24. The incorrect order among the following is/are
(A) LiH > NaH > KH (Thermal stability)
(B) $MgSO_4 > CaSO_4 > SrSO_4$ (Solubility in water)
(C) Li < Na < K < Rb (Density)
(D) $Be(OH)_2 > Mg(OH)_2 > Ca(OH)_2$ (Basic nature)

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match each set of hybrid orbital from List – I with the molecule or ion given in List - II

LIST–I		**LIST–II**	
P.	sp^2	1.	XeF_5^+
Q.	sp^3	2.	NO_3^-
R.	sp^3d	3.	NF_4^+
S.	sp^3d^2	4.	SF_2Cl_2
		5.	IOF_4^-
		6.	F_2SeO

(A) **P → 2; Q → 3; R → 4, 6; S → 1, 5**
(B) **P → 2; Q → 3, 6; R → 4; S → 1, 5**
(C) **P → 4; Q → 2; R → 3, 5; S → 6**
(D) **P → 6; Q → 3, 6; R → 4, 2; S → 1**

26. Match the resulting solution in List-I with one or more properties in List – II and choose the correct option. log2 = 0.3, log5 = 0.7. pK_a CH_3COOH = 4.74, pK_b NH_4OH = 4.74,

LIST–I		**LIST–II**	
P.	10 ml 0.1 M H_2SO_4 + 40 ml 0.1 M NH_4OH	1.	Acidic buffer solution
Q.	20 ml 0.1 M CH_3COOH + 16 ml 0.1 M NaOH	2.	Basic buffer solution
R.	50 ml 0.1 M CH_3COOH + 50 ml 0.1 M NaOH	3.	pH of the solution is 8.72
S.	10 ml 0.1 M NH_4OH + 20 ml 0.1 M NH_4Cl	4.	pH of the solution is 8.96
		5.	pH of the solution 9.26

(A) **P → 2, 4; Q → 1, 5; R → 3; S → 2**
(B) **P → 2, 3; Q → 1, 4; R → 3, 1; S → 2, 4**
(C) **P → 5; Q → 1; R → 3, 4; S → 2**
(D) **P → 2, 5; Q → 1; R → 3; S → 2, 4**

Space for Rough work

27. Match the half-life of the reaction in List-I with the integrated rate equation in List-II(a = initial concentration of reactant and a – x = concentration of reactant at time 't'.)

LIST–I		**LIST–II**	
P.	$t_{1/2} = \frac{0.586}{k}\sqrt{a}$	1.	$k = \frac{x}{t}$
Q.	$t_{1/2} = \frac{0.693}{k}$	2.	$k = \frac{1}{t}\left[\frac{1}{a-x} - \frac{1}{a}\right]$
R.	$t_{1/2} = \frac{a}{2k}$	3.	$k = \frac{2}{t}\left[\sqrt{a} - \sqrt{a-x}\right]$
S.	$t_{1/2} = \frac{1}{ka}$	4.	$k = \frac{2.303}{t}\log\frac{a}{a-x}$
		5.	$k = \frac{2}{t}\left[\frac{1}{\sqrt{a-x}} - \frac{1}{\sqrt{a}}\right]$

(A) **P → 5; Q → 4; R → 1; S → 2**
(B) **P → 3; Q → 4; R → 1; S → 2**
(C) **P → 2; Q → 3; R → 1; S → 2**
(D) **P → 3; Q → 2; R → 4; S → 5**

28. List-I contains carbides and List-II contains the type of carbide and their hydrolysis product.

LIST–I		**LIST–II**	
P.	Al_4C_3	1.	Covalent carbide
Q.	Mg_2C_3	2.	Ionic carbide
R.	CaC_2	3.	CH_4
S.	SiC	4.	C_3H_4
		5.	C_2H_2

(A) **P → 2, 3; Q → 2, 4; R → 2, 5; S → 1**
(B) **P → 1, 3; Q → 1, 4; R → 2, 5; S → 2**
(C) **P → 2, 4; Q → 1, 3; R → 1, 5; S → 5**
(D) **P → 2, 5; Q → 2, 4; R → 2, 3; S → 1**

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. A first order reaction has a rate constant 5×10^{-2} s^{-1}. The time required (in sec) for the reaction to reduce 20 gram of the reactant to 2 g is

30. 20 g of $CaCO_3$ is placed in a vessel of volume 12.315 litre at 1000 K. The following equilibrium was established.

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

K_P for the reaction at 1000 K is 0.3 atm. The mass of $CaCO_3$ in gram at equilibrium is

31. 100 ml of 0.1 M CH_3COOH, 50 ml 0.1 M HCl and 50 ml 0.1 M $Ba(OH)_2$ are mixed together. pK_a of CH_3COOH is 4.74. The pH of the resulting solution is

32. The mass of a dust particle is 10^{-12} g and its velocity is 10^{-4} cm/sec. The error in the measurement of velocity is 0.01%. The uncertainty in the measurement of position in A^o unit is $\left(\frac{h}{4\pi} = 0.527\times10^{-27} \text{ erg sec}\right)$

33. Vapour density of a metal chloride is 85.02. The equivalent weight of the metal is 7.01. The atomic weight of the metal is

34. In He^{2+} an electron undergoes transition from 3rd excited state of to 2nd energy level. The wavelength of the light emitted in A^o is $\left(\frac{1}{R} = 911.7A^o\right)$

35. In the thermal decomposition of NH_4NO_2, 50% of the compound decomposed in 34.65 min. The time required in minutes for the 90% decomposition of the compound is

36. The equilibrium constant (K_P) for a reaction at 27°C for a homogeneous gaseous reaction is 10^{-2}. The standard free energy change $\left(\Delta G^o\right)$ for the reaction in calorie is (R = 2 cal)

Space for Rough work

Mathematics

PART – III

SECTION – A

(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. For two positive integers m and n $\lim\limits_{x\to 0}\dfrac{\sqrt[m]{\cos x}-\sqrt[n]{\cos x}}{x^2}$, then which is true?

(A) m = 3, n = 2 it is $\dfrac{1}{6}$ (B) m = 5, n = 2 it is $\dfrac{3}{13}$

(C) m = 3, n = 2 it is $\dfrac{1}{12}$ (D) m = 5, n = 2 it is $\dfrac{3}{20}$

38. Let $f(x)=\dfrac{e^x}{1+x^2}$ and g(x) = f′(x), then

(A) g(x) has two local maxima and two local minima points
(B) g(x) has exactly one local maxima and one local minima point
(C) x = 1 is a point of local maxima of g(x)
(D) there is a point of local maxima for g(x) in the interval (–1, 0)

39. Let $f(x) = x^2 - 2|x|$ and $g(x)=\begin{cases}\min\{f(t), & -2\le t\le x, \quad -2\le x<0\\ \max\{f(t), & 0\le t\le x, \quad 0\le x\le 2\\ f(x) & x>2\end{cases}$, then g(x) is not differentiable at

(A) x = 0 (B) x = –1
(C) x = 2 (D) x = 1

Space for rough work

40. The function f: {0, 1, 2, ...} $\times$ {0, 1, 2, ...} $\to$ R satisfying

$$f(x, y) = \begin{cases} \frac{1}{2}\{f(x+1, y-1) + f(x-1, y+1)\} + 1 & \text{if } x \cdot y \neq 0 \\ 0 & \text{if } x \cdot y = 0 \end{cases}$$, then which is true

(A) f(3, 5) = 15 (B) f(5, 3) = 15
(C) f(9, 2) = 18 (D) f(2, 9) = 18

41. The equations of the normals to the curve $f(x) = \frac{x}{1-x^2}$ at the points where the tangents make angle of $\frac{\pi}{4}$ with the positive direction of x-axis are

(A) x + y = 0 (B) $x + y = \frac{\sqrt{3}}{2}$

(C) $x + y = 2\sqrt{2}$ (D) $x + y = \frac{-\sqrt{3}}{2}$

42. Let f be a function defined for all real x and let it satisfy the relation f(x + y) = f(x) + f(y) + xy(x + y) if f′(0) = –1, then
(A) f is differentiable for all real x
(B) f′ is differentiable for all real x
(C) f′(3) = 8
(D) f′ satisfies the relation $f'(x) + f'\left(\frac{1}{x}\right) = -f'(x)f'\left(\frac{1}{x}\right) \ \forall\ x \neq 0$

Space for rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Match the following List-I with List-II

LIST–I		**LIST–II**	
P.	If f: R $\to$ R is a differentiable function with the property that $\lim_{x\to\infty} f(x)$ exist and is finite, and if $\lim_{x\to\infty} x\cdot f'(x)$ exists, then this limit is equal to	1.	1
Q.	Let f and g be n-times continuously differentiable function in a neighbourhood of a point 'a', such that f(a) = g(a) = ln 2. $f'(a) = g'(a)$ ..... $f^{n-1}(a) = g^{(n-1}(a)$ and $f^n(a) \neq g^n(a)$, then $\lim_{x\to a} \frac{e^{f(x)} - e^{g(x)}}{f(x) - g(x)}$ is	2.	3
R.	Let f: [a, b] $\to$ R be a function continuous on [a, b] and differentiable on (a, b) if there exist C $\in$ (a, b) such that $\frac{f(b)-f(c)}{f(c)-f(a)} < 0$, then there always exist $\mu \in$ (a, b), then $f'(\mu)$ is equal to	3.	5
S.	Let a and b be positive real numbers, then $\int_a^b \frac{e^{\frac{x}{a}} - e^{\frac{b}{x}}}{x} dx$ is	4.	7
		5.	2
		6.	0

The correct option is:
(A) **P → 4; Q → 6; R → 2; S → 1**
(B) **P → 1; Q → 4; R → 2; S → 3**
(C) **P → 4; Q → 6; R → 5; S → 2**
(D) **P → 6; Q → 5; R → 6; S → 6**

Space for rough work

44. Match the following List-I with List-II

	LIST–I		LIST–II
P.	$\int_{-\pi}^{\pi} \frac{x^2}{1+\sin x+\sqrt{1+\sin^2 x}}dx$	1.	$\tan^{-1}\frac{2}{\pi}-\frac{\pi}{4}$
Q.	Evaluate $\int_{1/2}^{2} \frac{\tan^{-1}x}{x^2-x+1}dx$	2.	0
R.	$\int_{0}^{\pi/2} \frac{x\cos x-\sin x}{x^2+\sin x}dx$	3.	$\tan^{-1}\frac{3}{\pi}-\frac{\pi}{4}$
S.	$\int_{-1}^{1} \frac{\sqrt[3]{x}}{\sqrt[3]{1-x}+\sqrt[3]{1+x}}dx$	4.	$\frac{\pi\sqrt{3}}{18}$
		5.	$\frac{\pi^3}{3}$
		6.	$\frac{\pi^2\sqrt{3}}{18}$

The correct option is:
(A) **P → 5; Q → 6; R → 2; S → 1**
(B) **P → 1; Q → 4; R → 2; S → 3**
(C) **P → 5; Q → 6; R → 1; S → 2**
(D) **P → 4; Q → 2; R → 3; S → 1**

Space for rough work

45. Match the following List-I with List-II

LIST–I		LIST–II	
P.	f: R $\to$ R; $f(x)=\left[\left\{\left\|e^{x}\right\|\right\}\right]$ (where [.] denotes the greatest integer function, and {.} denotes fractional part function)	1.	one-one
Q.	Let f: R $\to$ R defined as $f(x)=e^{\text{sgn}(x)}+e^{x^2}$ (where sgn x denotes signum function of x, then f(x))	2.	many one
R.	f: R $\to$ R; $f(x)=\left[x+\frac{1}{2}\right]+\left[x-\frac{1}{2}\right]+2[-x]$ (where [.] denotes the greatest integer function)	3.	into and periodic
S.	f: [–1, ∞) $\to$ (0, ∞) defined as $f(x)=e^{x^2+\|x\|}$, then f(x) is	4.	onto
		5.	periodic
		6.	constant

The correct option is:
(A) **P $\to$ 4; Q $\to$ 6; R $\to$ 2; S $\to$ 1**
(B) **P $\to$ 2; Q $\to$ 2; R $\to$ 5; S $\to$ 2**
(C) **P $\to$ 4; Q $\to$ 6; R $\to$ 5; S $\to$ 2**
(D) **P $\to$ 2; Q $\to$ 2; R $\to$ 3; S $\to$ 1**

Space for rough work

46. Match the following List-I with List-II

LIST–I		LIST–II	
P.	Area bounded by the curve $y - \sin^{-1} x = x - x^2$, and the x-axis is equal to	1.	0
Q.	The area bounded by $y = x^2 - 3$ and the line $y = ax + 2$ attains its minimum value than the parameter 'a' is equal to	2.	1
R.	The area of the region represented by the expression $\sqrt{2} \le \lvert x+y\rvert + \lvert x-y\rvert \le 2\sqrt{2}$	3.	$\pi - 1$
S.	If k is a positive number and the area of the region bounded by the curves $y = x - kx^2$ and $ky = x^2$ attains its maximum value, then k is equal to	4.	6
		5.	2
		6.	3

The correct option is:
(A) **P → 3; Q → 6; R → 2; S → 1**
(B) **P → 1; Q → 4; R → 2; S → 3**
(C) **P → 4; Q → 6; R → 5; S → 2**
(D) **P → 3; Q → 1; R → 4; S → 2**

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. If limit $\lim\limits_{n\to\infty} n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)}\left(1^1 \cdot 2^2 \cdot 3^3 \ldots\ldots n^n\right)^{\frac{1}{n^2}}$ = L, then –ln L is _____

48. The positive value of 'a' such that the parabola $y = x^2 + 1$, bisects the area of the rectangle with vertices (0, 0), (a, 0), $(0, a^2 + 1)$, $(a, a^2 + 1)$, then $\frac{a}{2\sqrt{3}}$ is _____

Space for rough work

49. Let A – rational number between (0, 1), and a function f: A $\to$ R has the property that for all x $\in$ A $f(x)+f\left(1-\frac{1}{x}\right)=\ln|x|$, then the value and $\frac{2001}{2014}e^{f(2007)}$ is _____

50. The value of $\frac{k}{2}$ which minimizes $F(k)=\int_0^4|x(4-x)-k|dx$ _____

51. For x a real number, let f(x) = 0 if x < 1 and f(x) = 2x – 2 if x $\geq$ 1. If number of solutions to the equation $f(f(f(f(x))))=x$ is n, then $\frac{n}{5}$ is equal to _____

52. For any real $\lambda \geq 1$ denote by $f(\lambda)$ the real solution to the equation $x(1 + \ln x) = \lambda$, then $\frac{1}{2}\lim_{\lambda\to\infty}\frac{f(\lambda)}{\left(\frac{\lambda}{\ln\lambda}\right)}$ is _____

53. $\int(x^6+x^3)(x^3+2)^{\frac{1}{3}}dx = \lambda(x^6+2x^3)^{\frac{4}{3}}+c$, then 2λ is equal to _____

54. The differentiable function F: R $\to$ R satisfies F(0) = –1 and $(F'(x))=\sin(\sin(\sin(\sin(x))))\cdot\cos(\sin(\sin(x)))\cdot\cos(\sin(x))\cdot\cos(x)$. Then $-\frac{F(\pi)}{2}$ is _____

Space for rough work